

Urban Visual Genes: Discovering Reproducible and Interpretable Visual Primitives from Street-Level Imagery

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Abstract

Street-view imagery supports measurement of the built environment at scale, but most existing pipelines either predict a predefined label or return dense features that are difficult to interpret. We introduce Urban Visual Gene, a framework that decomposes patch-level representations from a frozen DINOv3 vision transformer into sparse, recurring visual primitives using a BatchTopK sparse autoencoder (SAE). Our main experiment uses 14,400 directional street-view images from 12 globally distributed cities, yielding 11.29 million image patches. A 1,024-dimensional dictionary with eight active genes per patch reconstructs the source representation with approximate cosine similarity 0.849 while retaining broad dictionary utilization. Across five independent training runs, we identify 875 gene families present in at least four runs, including 738 recovered in all five. A language-free multiview analysis organizes the dictionary into 32 coarse and 64 fine branches using decoder, encoder, spatial-position, city-distribution, and activation-context evidence. Gene profiles predict held-out road class with mean balanced accuracy 0.622 under spatial cross-validation, compared with 0.388 for acquisition and image-quality nuisance variables alone. Finally, an open-source Qwen3-VL model produces structured concept descriptions for a stratified pilot of 128 genes. These descriptions are preliminary: their semantic fidelity has not yet been independently validated. Together, the results establish a reproducible basis for interpretable, comparative “visual genomes” of cities while exposing important challenges in redundancy, nuisance features, calibration, and semantic evaluation.

1 Introduction

Street-level imagery offers a human-scale view of cities and has been used to study urban form, transportation, vegetation, health, socioeconomic conditions, and perception [1]. Yet common analytical approaches depend on fixed semantic segmenters or supervised prediction targets. Such methods are useful when the relevant ontology is known in advance, but they cannot easily discover recurring visual structure outside that ontology. Conversely, foundation-model embeddings capture rich visual information but do not directly expose a compact, human-auditable vocabulary.

We ask whether the visual character of cities can instead be represented through a shared set of learned primitives. We call a recurring sparse feature a *visual gene*, and the distribution and co-occurrence of genes within a city its *visual genome*. The biological metaphor is deliberately limited: genes here are learned components of a visual representation, not causal or immutable attributes of a city.

Our framework applies a sparse autoencoder to dense, patch-level DINOv3 representations [8]. Patch observations preserve localized evidence, while a shared encoder makes a feature independent of which spatial patch contains it. BatchTopK sparsity [2] constrains the average number of active latents and permits different patches to use different numbers of features. We then evaluate reconstruction, utilization, reproducibility across random initializations, statistical structure, external validity, and nuisance sensitivity. A vision-language model (VLM) supplies provisional natural-language names for a stratified subset, following the broader idea that SAE features can be translated into testable hypotheses [7].

Our contributions are:

1. a patch-level SAE formulation for discovering a cross-city vocabulary of urban visual primitives;
2. a reproducibility analysis that matches features across five independently trained dictionaries using both activation and decoder similarity;
3. a language-free hierarchy and evidence audit that separates stable, city-specific, rare, redundant, and nuisance-related components;
4. spatially blocked external-validity and nuisance-control experiments; and
5. an open-source VLM pipeline for structured interpretation of visual genes, with held-out fidelity evaluation explicitly separated from label generation.

2 Related Work

Street-view imagery for urban measurement. Street-view imagery is now a standard source for urban analytics, spanning physical elements and perceived qualities [1, 5]. Much of this literature maps images to predefined outcomes. Our objective is complementary: to discover a reusable vocabulary before selecting a downstream target.

Self-supervised visual representations. Vision transformers learn patch tokens that support dense visual correspondence [4, 3]. DINOv3 improves the quality and stability of dense self-supervised features at scale [8]. We treat these features as the observation space to be decomposed rather than as an opaque endpoint.

Sparse autoencoders and feature interpretation. SAEs seek an overcomplete representation in which each observation is reconstructed from a small number of learned directions. TopK and BatchTopK variants directly control sparsity [6, 2]. Natural-language interpretation can make learned features accessible, but a plausible label is not evidence of fidelity. Recent work therefore separates label generation from tests of whether that label predicts feature activation [7]. We adopt the same distinction for visual features.

3 Data

The current scientific analysis covers 12 cities: Hong Kong, Singapore, Amsterdam, Cape Town, Paris, São Paulo, Mexico City, Sydney, Jakarta, Dhaka, New Delhi, and Manila. We sample 1,200 directional images per city, for 14,400 images total. Each image is represented by a grid of DINOv3 patch tokens, giving 11,289,600 patch observations. The balanced city sample prevents the largest source collection from dominating dictionary learning.

The broader acquisition pipeline currently contains data infrastructure for 110 cities and approximately 40.76 million panoramas (about 163 million directional views), but those images are *not* all used in the experiments reported here. They provide a path for future global-scale encoding. **[TODO: Add collection dates, road-sampling interval, panorama deduplication rules, image resolution, licensing, and a formal ethics/privacy statement.]**

For external validation, we aggregate four headings to the panorama level for 3,600 locations (300 per city). Road labels are derived from the nearest OpenStreetMap highway class. Building coverage, count, and distance targets use Global Building Atlas footprints within 100 meters; building validation is presently available for four cities.

4 Method

4.1 Patch representations

Let $x_i \in \mathbb{R}^d$ denote a DINOv3 patch embedding. We treat each patch as a separate observation and share the SAE parameters across all positions. This avoids learning a separate encoding weight for, for example, the same bus appearing at two neighboring patch indices. The backbone is frozen throughout.

4.2 BatchTopK sparse autoencoder

The encoder produces preactivations

$$h_i = W_{\text{enc}}x_i + b_{\text{enc}}. \quad (1)$$

BatchTopK retains the largest kB positive activations across a batch of B observations, producing sparse code z_i . The decoder reconstructs

$$\hat{x}_i = W_{\text{dec}}z_i + b_{\text{dec}}. \quad (2)$$

We minimize cosine reconstruction loss,

$$\mathcal{L}_{\text{recon}} = 1 - \frac{x_i^\top \hat{x}_i}{\|x_i\|_2 \|\hat{x}_i\|_2}. \quad (3)$$

The working model has width 1,024 and target sparsity $k = 8$, and is trained for 60 epochs. We call each decoder direction and its associated sparse activation a visual gene.

4.3 Model selection

We compare dictionary widths $\{1024, 2048\}$ and sparsity levels $k \in \{4, 8, 16\}$. Selection is not based on reconstruction alone: a useful model must also retain dictionary utilization, manageable redundancy, and sparse local explanations. We therefore adopt W1024/K8 as a working compromise.

4.4 Cross-seed reproducibility

We train five independent W1024/K8 dictionaries with seeds 11, 23, 37, 53, and 71. For each pair of runs, candidate genes must have decoder cosine similarity at least 0.7, activation similarity at least 0.7 on 250,000 anchor patches, and support on at least 25 anchors. We join accepted matches into gene families and call a family stable when it appears in at least four seeds.

4.5 Language-free hierarchy and evidence audit

To avoid imposing semantic labels on the statistical structure, we build five gene views: encoder weights, decoder weights, spatial activation maps, city activation profiles, and activation context. A rank-fused graph followed by spectral embedding and Ward clustering yields a nested hierarchy. Model-selection diagnostics favor 32 coarse and 64 fine branches. Each gene receives evidence descriptors for stability, support, specificity, nuisance risk, and redundancy.

4.6 VLM interpretation pilot

For 128 genes selected by a stratified audit, Qwen3-VL-8B-Instruct receives six pairs of original images and activation overlays. Only top and cross-city exemplars are used for label generation. The model returns a concept phrase, description, concept type, alternative hypothesis, confidence, and position-artifact flag. Random-positive and borderline-positive examples are held out for future fidelity testing. Because the generator’s confidence is not calibrated and no independent judge has yet been run, we report these labels only as preliminary annotations.

Table 1: Width and TopK ablation. Reconstruction cosine is approximated as one minus final cosine loss. “Used” counts genes passing the cross-city prevalence analysis in the corresponding run.

Width	k	Recon. cosine	Used genes	Final loss
1024	4	0.809	785	0.191
1024	8	0.849	929	0.151
1024	16	0.881	898	0.119
2048	4	0.815	1035	0.185
2048	8	0.858	1328	0.142
2048	16	0.891	1433	0.109

Stable visual-gene activations across five SAE seeds

One shared reference per family - identical colormap, scale, threshold, and $\alpha = 0.58$ across all activation panels

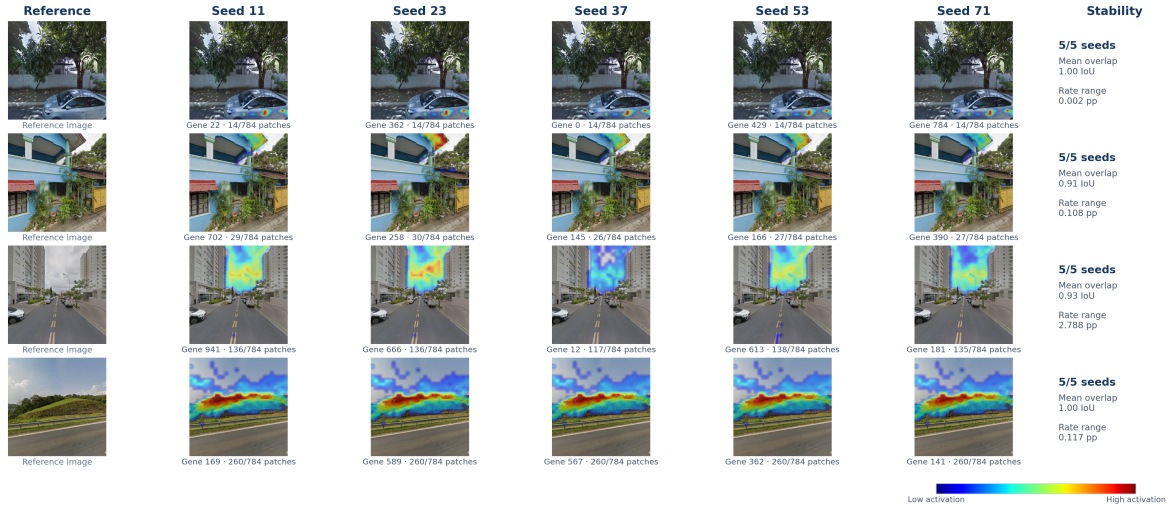


Figure 1: Representative cross-seed gene families. Features are matched using decoder and held-out activation similarity rather than numeric gene IDs.

5 Experiments and Results

5.1 Reconstruction–sparsity trade-off

As expected, reconstruction improves with larger width and larger k (table 1). W2048/K16 obtains the best reconstruction, but requires a larger vocabulary and less sparse explanations. W1024/K8 retains a reconstruction cosine of 0.849 with eight target activations per patch and is used below.

In the final BatchTopK run, all 1,024 dimensions activate somewhere, while 896 pass a city-level prevalence threshold of 0.0005. Of the full dictionary, 79 genes are prevalent in all 12 cities, 130 in 9–11 cities, 184 in 6–8, 257 in 3–5, 121 in two, and 125 in one; 128 fall below the city prevalence threshold. These counts depend on the stated threshold and should not be read as intrinsic biological-like categories.

5.2 Reproducibility

All five 60-epoch runs converge to final loss between approximately 0.154 and 0.155. Matching produces 8,175 accepted pairwise correspondences and 875 stable families: 738 are recovered in all five seeds and 137 in four. Thus, a substantial majority of the dictionary participates in reproducible cross-run structure, although matching does not prove semantic purity.

5.3 Statistical taxonomy

The fused graph is connected and supports 32 coarse and 64 fine branches. Evidence roles include 251 stable structured genes, 107 city-specific components, 85 rare stable components, 234 weakly structured components, 340 unstable components, two position components, and five quality components. A stricter evidence audit finds 73 robust structured and 13 robust specific genes, while identifying 110 redundant variants and seven nuisance components. Strict equivalence grouping suggests an effective dictionary size of 932. These results show that the learned dictionary is neither wholly semantic nor wholly noise: it mixes robust visual structure with unstable, redundant, and acquisition-related directions.

5.4 External validity and nuisance controls

Under five contiguous within-city spatial folds, gene profiles predict the nearest road class with mean balanced accuracy 0.622 across 12 cities. This exceeds the class prior (0.336) and nuisance-only features (0.388), and is comparable to dense DINO features (0.616). Adding nuisance features to genes gives 0.620, suggesting that the road signal is not explained by capture year, season, brightness, blur, sky fraction, or QC flags alone.

For building targets in four cities, gene profiles show moderate average rank association: Spearman $\rho = 0.462$ for building coverage, 0.497 for log building count, and 0.507 for distance to the nearest building. However, their mean held-out R^2 values are strongly negative, indicating poor calibration and spatial generalization under the current regression. Dense DINO features perform better in R^2 . We therefore treat building results as evidence of ordering information, not accurate metric prediction.

For city classification over 3,600 panoramas, nuisance-only variables achieve 0.247 balanced accuracy, compared with 0.953 for gene profiles and 0.967 for dense DINO. Residualizing nuisance variables changes gene accuracy only modestly (0.949). Median cross-validated nuisance R^2 across genes is 0.010 (90th percentile 0.062), although the most nuisance-predictable gene reaches 0.462. Nuisance control is thus important at the feature level even when aggregate city prediction remains strong.

5.5 Preliminary VLM interpretations

Qwen returns valid structured JSON for all 128 pilot genes. The generated concept types comprise 68 parts, 22 objects, 16 scenes, 10 artifacts, six textures, five materials, and one layout. Examples include “paved road surface” (gene 405), “man-made structure” (gene 120), “clear blue sky” (gene 681), “windowed facade” (gene 554), and “delivery truck” (gene 581). The average self-reported confidence is 0.956, but the model marks no feature as a positional artifact even though the statistical audit identifies position-related cases. This discrepancy demonstrates why self-reported confidence is insufficient and motivates the held-out fidelity experiment.

6 Discussion

The experiments support three conclusions. First, a sparse dictionary can retain most of the local DINOv3 representation while exposing a compact set of activated features. Second, much of this structure is reproducible across random initializations, which is a stronger claim than interpretability from a single training run. Third, gene distributions contain externally relevant urban-form signal beyond simple acquisition and image-quality covariates.

At the same time, “visual gene” should not imply one feature equals one clean human concept. Some dimensions are broad, redundant, unstable, positional, or related to image quality. The VLM pilot makes the output easier to browse but also reveals a failure mode: fluent, high-confidence descriptions can miss statistically diagnosed artifacts. A defensible release should therefore expose evidence scores and counterexamples alongside every natural-language label.

7 Limitations and Next Steps

- The reported SAE is trained on 12 cities; claims about the 110-city corpus require global encoding and geographically held-out evaluation.

- The current main experiment treats patches as observations, but a controlled comparison against global embeddings, four-view means, separate directional views, and concatenated views remains unfinished because of cluster-environment failures.
- The VLM pilot lacks independent fidelity scores. We will ask a separate judge (and a blinded human subset) whether generated concepts distinguish held-out positive from matched negative examples.
- Gene matching uses fixed similarity thresholds. Sensitivity analysis and uncertainty intervals are needed.
- External building prediction is poorly calibrated under spatial folds; downstream interpretation should emphasize rank association until improved.
- Google Street View sampling, geographic coverage, temporal variation, and platform-specific artifacts can bias cross-city comparisons.

8 Conclusion

Urban Visual Gene turns dense patch representations into a sparse, auditable vocabulary for comparative urban imagery. The present evidence establishes reconstruction, cross-seed reproducibility, statistical organization, and partial external validity. The next scientific priority is not simply scaling the dictionary, but validating which individual genes have faithful semantic interpretations and remain robust across cities, time, position, and acquisition conditions.

Data and code availability. An interactive research dashboard is available at <https://cehougis.github.io/urban-visual-gene/>. [TODO: Add an archival code release, model checkpoint license, street-view redistribution statement, and DOI.]

Acknowledgments. [TODO: Add collaborators, funding, Savio/BRC acknowledgement, and data-provider acknowledgments.]

References

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A Current project status

As of September 2026, the main W1024/K8 model, width/sparsity ablations, five-seed matching, statistical taxonomy, external-validity analysis, nuisance controls, and 128-gene Qwen labeling run are complete. The controlled observation-unit ablation and independent VLM fidelity evaluation remain incomplete. This appendix is a draft status record and should be removed or updated before submission.

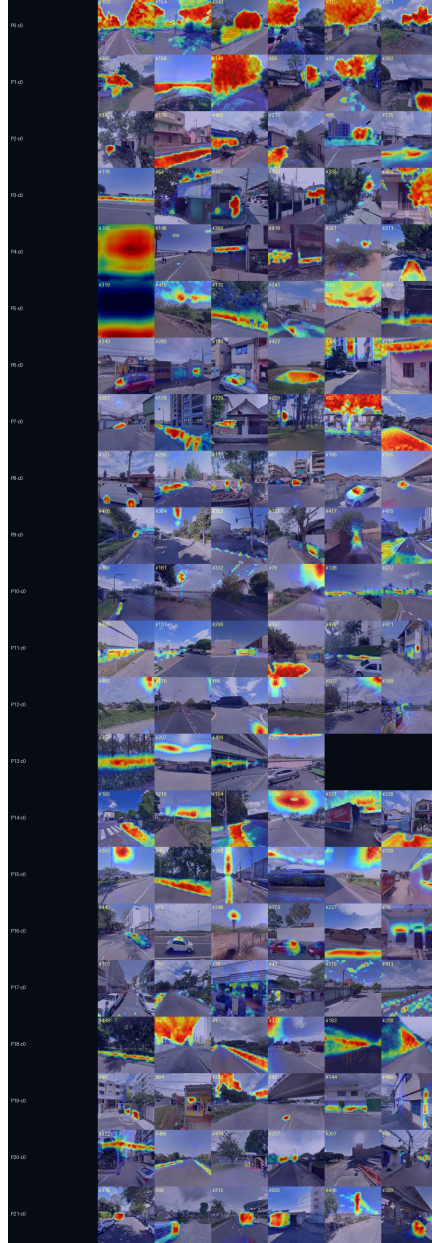


Figure 2: Visual-gene organization. The displayed semantic categories are an interpretive view; quantitative hierarchy selection itself uses no language labels.